

1	2
3	4

1 – 70p_ASTRAL_partition_entropy
2 – 70p_ASTRAL_uce
3 – 70p_ghost
4 – 70p_partition_entropy

>=98 (SH-aLRT/UFBoot2) or >=0.99 (ASTRAL)

>=80–97.9 (SH-aLRT) or >=95–97 (UFBoot2) or >=0.95–0.98 (ASTRAL)

>=50–79.9 (SH-aLRT) or >=50–94 (UFBoot2) or >=0.5–0.95 (ASTRAL)

<50 (SH-aLRT/UFBoot2) or <0.5 (ASTRAL)

monophyly not supported

not computed

A phylogenetic tree of Lucanidae species, rooted at the top left. The tree is annotated with support values (SH-aLRT/UFBoot2) and node markers (ASTRAL). The species names and their identifiers are listed on the right side of the tree. The tree shows a clear relationship between the species, with some groups being more closely related than others. The support values are indicated by the size and shading of the nodes. The node markers are indicated by the presence or absence of a cross in the node. The tree is rooted at the top left, with the root node having a support value of 100/100. The tree branches out to the right, with the species names and their identifiers listed on the right side of the tree. The species names are: *Lucanus capreolus* COL3897, *Xylonychus* sp COL066, *Trichaulax* sp COL1050, *Anoplostethus* sp COL892, *Sarophorus costatus* ST007, *Dicranocara deschodti* STL10269, *Endroedyolus paradoxus* STL10049, *Sisyphus* –*Neosisyphus*– *calcaratus* STL7, *Sisyphini* sp morpho1 STL18, *Epactoides femoralis* STL10145, *Panelus* sp3 STL51, *Gyronotus pumilus* STL10003, *Chalconotus convexus* STL10294, *Kheper nigroaeneus* STL10036, *Drepanopodus proximus* STL10001, *Nanos* sp1 STL10147, *Nanos* sp2 STL10153, *Cambefortantus pauliani* STL27, *Apotolamprus aff ambohitsitondronensi* STL10140, *Nanos aff rubrosignatus* STL10142, *Nanos* sp7 STL5005, *Nanos* sp10 STL10206, *Nanos dubitatus* STL5001, *Metacatharsius* sp1 STL10007, *Metacatharsius opacus* STL10017, *Macroderes* sp1 STL10328, *Onitis alexis* STL10041, *Onthophagus* *Serrophorus seniculus* STL10110, *Liatongus* sp STL10290, *Cytochirus* sp1 STL10112, *Euoniticellus fulvus* STL10132, *Helictopleurus* sp group *semivirens* STL10214, *Helictopleurus neuter* STL10327, *Helictopleurus fasciolatus pseudofasciolatus* STL10333, *Helictopleurus furcicornis* STL10169, *Helictopleurus coruscus* STL10209, *Helictopleurus aurocupreus* STL5025, *Onthophagus sl batillifer* STL10301, *Onthophagus sl cf quadrimaculatus* STL10046, *Arachnodes* sp4 STL5016, *Arachnodes* sp STL10216, *Arachnodes* sp. morpho 1 STL45, *Arachnodes* sp6 STL10200, *Onycothechus tricornigeris* STL10225, *Copris sl diversus* STL10245, *Litocopris muticus* STL10027, *Copris urus* STL10024, *Copris gilleti* STL10028, *Temnoplectron lewisensis* Temlew UCE, and *Thyregeis relictus* Thyrel UCE. The support values are indicated by the size and shading of the nodes. The node markers are indicated by the presence or absence of a cross in the node. The tree is rooted at the top left, with the root node having a support value of 100/100. The tree branches out to the right, with the species names and their identifiers listed on the right side of the tree.